

2024



Centrica Drives Cloud-First Strategy While Delivering Greater Value to Customers with AWS

By collaborating with AWS experts and investing in employees' AWS skills, Centrica is increasing cloud usage, modernizing environments, and enhancing customer experiences.

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90%

reduction in cost to serve

Fro

accel

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600,000

automated annual software deployments annually

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internet or things devices managed

From 60 per week to 9 per month

Overview

Energy solutions and services company [Centrica](#) is building cloud-native solutions while ensuring best practices around compliance and security with AWS. Collaborating with AWS, Centrica is upskilling its employees, utilizing AWS expertise to migrate core platforms, and accelerating the development of innovative, cloud-native solutions for sustainable energy services to customers.

As a result, the company has reduced the cost to serve customers, increased performance, and enhanced developer productivity through new approaches. Furthermore, Centrica is promoting greater workforce sustainability by facilitating the improvement and acquisition of AWS skills among its employees.

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Opportunity | Driving Sustainability and Cloud-First Strategy

Based in the UK and Ireland, Centrica is an energy services and solutions provider serving more than 10 million residential and business customers. Centrica, with a strong focus on sustainability, aims to achieve net-zero operations by 2045 and assists customers in reaching the same goal by 2050.

In pursuit of its net-zero targets, Centrica is shifting from fossil fuels to cleaner energy sources such as wind and solar. The organization has created services such as British Gas Net Zero to help customers achieve their net-zero goals.



technologies like heat pumps and install charging points for electric vehicles. It has a subsidiary company called Hive that produces a range of app-controlled smart-home products, including thermostats and light switches, empowering consumers to have more control over their energy consumption.

To help reduce its own carbon footprint and enhance the efficiency of its IT infrastructure, Centrica began migrating IT systems in 2018 from on-premises data centers to Amazon Web Services (AWS). In 2020, the company had migrated 90 percent of its IT environment to the cloud, and by 2023, it had restructured 40 percent of its workloads to operate natively in the cloud. Centrica aims to close its last on-premises facility and be 100 percent cloud native by 2027.

In line with this strategy, Centrica is working with AWS experts to optimize its environments from both cost and performance perspectives and to migrate its remaining workloads to AWS. In addition, it is developing an in-house team of cloud experts with AWS support to increase cloud skills, improve productivity, and reduce reliance on third parties.

Solution | Building Cloud Skills and Working with AWS Experts

Centrica leveraged the combined expertise of AWS and received instructional support to advance its cloud projects, benefiting from the contribution of [AWS Professional Services](#) and [AWS Training and Certification](#) to ensure the success of these initiatives.

In the past, Centrica's team primarily managed third-party providers and on-premises infrastructure, lacking the skills required for cloud-native development. Recognizing this gap, Centrica engaged AWS to train its engineers, empowering them to lead cloud projects and navigate the cloud confidently. James Boswell, director of cloud at Centrica, says, "We wanted our cloud teams at Centrica to be future ready."

AWS training played a pivotal role in the transformation at Centrica, with the company making substantial investments to upskill its workforce. In 2022, it dedicated 2,456 hours to provide [AWS Classroom Training](#) to 167 individuals. This comprehensive training program encompassed 11 trainer-led AWS Classroom courses with the focus on [AWS Cloud Practitioner Essentials](#), [Architecting on AWS](#), and [Developing on AWS](#). Centrica's training strategy extended beyond classroom learning to a Grow Your Own program that worked in tandem with this initiative used [AWS Skill Builder](#), leveraging its free resources for self-paced learning. It also included hands-on training, including coding challenges, for an immersive learning experience. In addition, Centrica joined the AWS Cloud Academy for practical experience in cloud engineering and set up a Cloud Council, which is supported by AWS.



Centrica's approach to training is tailored to meet the diverse needs of its teams and upcoming projects, ensuring its workforce is well-prepared for specific challenges, such as migrating from legacy platforms like SAP. Boswell emphasizes its commitment to personalizing the training experience, enhancing accessibility through an intranet site for personnel to select programs that align with their roles and career goals. "We worked with AWS to personalize the training experience for different roles and teams, rather than taking a one-size-fits-all approach," he says.

The availability of comprehensive AWS training at Centrica has not only increased employee engagement but has also reduced attrition rates, as employees feel valued with accessible, relevant training opportunities.

Thomas Whiffing, a program graduate and software engineer at Centrica, is a testament to the success of these initiatives. "These programs have ensured that we build services following AWS best practices, ultimately enhancing the customer experience," he attests.

Whiffing and his colleagues collaborate closely with AWS experts to develop new customer services and deploy cloud-native solutions. In 2023, the collaboration between Centrica and AWS led to the creation of a data analysis environment to assess the performance of a customer-facing chatbot, expertly integrating [Amazon AppFlow](#), [AWS Glue](#), and [AWS Lambda](#) serverless services. This not only optimized costs but also established best practices from the project's outset.

Centrica's commitment to cultivating cloud expertise extends to hiring entry-level cloud talent, facilitated through the [AWS re/Start](#), a cohort-based workforce development training program that prepares individuals for careers in the cloud and connects them to potential employers. This initiative provides individuals without an IT background the opportunity to embark on a cloud career within Centrica, as noted by Boswell: "AWS re/Start has been brilliant at giving people without an IT background the chance to launch a cloud career here at Centrica."

Outcome | Advancing Operational Efficiency and Customer Experience

Through its collaboration with AWS and improved AWS proficiency of technical teams, Centrica achieved significant improvements in system stability and resilience, reducing priority incident numbers from 60 per week to an impressive 9 per month.

Centrica's commitment to operational excellence is evident in its ability to execute 600,000 fully automated software changes annually without any incidents. The company's Internet of Things platform successfully scaled to manage over 3 million devices, demonstrating its capacity to handle substantial IoT demands.

Centrica's transition to a cloud-native environment significantly expedited updates for its billing system, reducing the time required from 3–6 months to mere weeks. In addition, with AWS support, Centrica has built an orchestration layer on AWS to simplify integrations with a number of revenue-generating aggregators and its system of records.

When energy prices recently surged, Centrica quickly scaled its cloud-based meter reading system to handle 500,000 readings in a single day, ensuring customers were well-served and protected. "AWS helped us configure and scale rapidly to prevent any customer harm," says Boswell. Furthermore, Centrica optimized its AWS environments, reducing the cost to serve the Hive platform by 90 percent.

Beyond its impact on customer service, Centrica's collaboration with AWS has led to increased engagement and reduced staff turnover. The company utilizes cloud-based training and retraining programs to



emphasize its commitment to its workforce. Experienced developers have gained confidence in working with artificial intelligence (AI) and machine learning (ML) services on AWS.

These skilled developers are integrating services like [Amazon SageMaker](#) into Hive to help customers predict heating costs and better manage consumption. “Our engagements with AWS are providing us with the confidence to incorporate newer technologies that positively impact our customers,” explains Boswell.

Through the AWS re/Start program, Centrica promotes diversity among its team of software engineers. Boswell notes, “Gender balance is important to us as a business so it’s great that our engineering team is 33 percent female, a figure we hope to increase over time.”

About Centrica

Centrica is an international energy services and solutions provider that supplies electricity and gas. The organization’s trusted brands include British Gas, Bord Gáis Energy, and Centrica Business Solutions.

AWS Services Used

AWS Training and Certification

Learn from AWS experts. Advance your skills and knowledge. Build your future in the AWS Cloud.

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AWS Professional Services

The AWS Professional Services organization is a global team of experts that can help you realize your desired business outcomes when using the AWS Cloud.

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AWS re/Start

AWS re/Start is a cohort-based workforce development training program that prepares individuals for careers in the cloud and connects them to potential employers.

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EUROPE, MIDDLE EAST, & AFRICA



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Electra, opérateur propriétaire de bornes de recharge ultra-rapide pour véhicules électriques, a choisi AWS dès ses débuts pour assurer la fiabilité, la scalabilité et la sécurité de son infrastructure. En tant que CPO (Charge Point Operator), Electra se distingue par l'installation de bornes ultra-rapides dans les zones urbaines et périurbaines, permettant aux utilisateurs de



2024

UNITED STATES



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